

Normal Image Guided Segmentation Framework for Unsupervised Anomaly Detection

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Abstract—Unsupervised anomaly detection is required to detect/segment anomalous samples/regions that deviate from the normal pattern while learning only through the normal sample category. Towards this end, this paper proposes a novel framework for anomaly detection by introducing normal images as guidance called Normal Image Guided Segmentation Framework (NIGSF). It consists of a Normal Guided Network (NGN) and a Saliency Augmentation Module (SAM). NGN constructs the contrast set, which is a candidate set for extracting normal sample features. Then, a normal feature extractor is developed to extract detailed and complete features containing normal semantic information as guidance features. Meanwhile, the guidance feature fusion module is introduced to realize normal semantic guidance in the feature space, and then the segmentation module discriminates the features that are different from the normal guidance features as anomalies. SAM aims to generate forged anomaly samples utilizing available normal samples. It introduces saliency maps and random Perlin noise to generate saliency Perlin noise maps and then to generate diverse forged anomaly samples. Extensive experiments are conducted to evaluate the performance of NIGSF on three anomaly detection benchmark datasets. The results demonstrate the effectiveness of each proposed module and the superiority of the proposed method. Specifically, NIGSF outperforms the runner-up by 5.4% in terms of anomaly segmentation AP metric.

Index Terms—Anomaly detection, contrast set, normal guidance, saliency augmentation, feature guidance.

I. INTRODUCTION

UNSUPERVISED anomaly detection is a crucial task in computer vision, which includes two parts: anomaly classification and anomaly segmentation [1], [2], [3], [4], [5]. The former aims to identify samples as normal or anomalous, whereas the latter strives to locate anomalous pixels. However, anomaly detection is different from general detection and segmentation tasks. Due to the wide distribution of

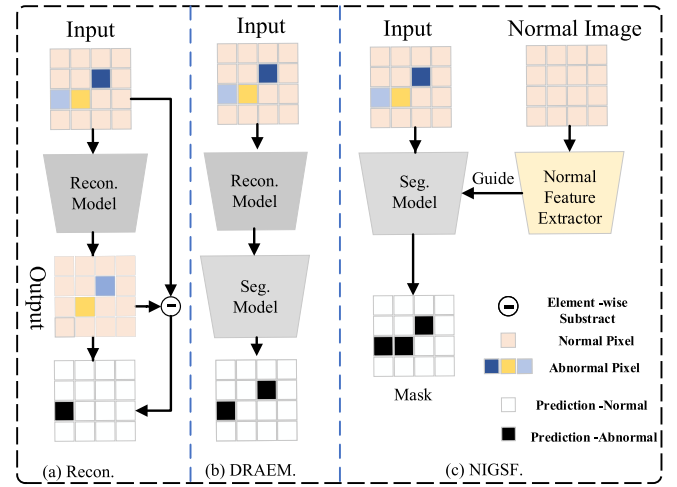


Fig. 1. Comparison of different unsupervised anomaly detection methods. (a) “Recon.” (Reconstruction-based method [6], [19]). (b) DRAEM [20]. (c) The proposed framework (NIGSF). NIGSF takes full advantage of the similarity of normal samples and directly utilizes normal image to guide the segmentation of abnormal regions.

anomaly events and the difficulty of collecting all anomaly categories [6], [7]. Therefore, unsupervised anomaly detection that relies only on normal sample modelling becomes an important task. This task holds great promise in industrial contexts [1], [2], [8], [9], [10], [11], medical diagnosis [12], [13], and video surveillance [14], [15], [16], [17], [18], for example, by modeling normal industrial product images to detect abnormal types of products and get rid of manual quality control. Training only through normal samples poses a huge challenge of unsupervised anomaly detection. To solve this challenge, the relationship between normal images must be fully explored.

Recent developments in anomaly detection have explored a plethora of methods to address its inherent challenges [21], [22], [23], [24]. As depicted in Figure 1 (a), popular reconstruction-based methods [6], [25], [26] utilize an autoencoder for recovering normal samples with low error, while recovering “unlearn” abnormal samples are expected to yield higher reconstruction error. This reconstruction error, representing the anomaly score of an image, alongside the pixel-level error forming the anomaly segmentation map, are the principal insights of these methods [3], [4]. However, these models face limitations as they cannot filter all anomalies, especially when actual anomalous samples for training are

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insufficient or when these anomalies closely resemble normal samples. Figure 1 (b) illustrates the DRAEM segmentation method [20], which anticipates training a segmentation model for direct anomaly segmentation. The insight of DRAEM expects the segmentation model to learn the representation of anomalies and then segmentation. However, anomalous features are diverse, which results in unlearned anomaly types that may still be misclassified as normal. Under the traditional anomaly detection paradigm, a significant limitation of existing methods is that they aim to learn as much as possible about the representation of all anomaly types and then filter or segment them accordingly. Therefore, there is an urgent requirement to develop a novel anomaly detection paradigm to overcome this challenge. Moreover, some methods [20], [27] have been proposed to overcome the imbalance challenge between positive and negative samples associated with the dearth of anomalous samples available for training. CutPaste [27] generates forged anomalous samples by randomly cropping thin rectangular areas from normal samples and then uses a classifier to detect the anomalous samples. Meanwhile, anomaly segmentation maps are inferred by the classifier's gradient. Nonetheless, the forged anomalous regions only comprise part of the normal sample and are rectangular in shape, which causes the forged anomalies to be too simplistic. DRAEM [20] is developed based on random noise [28] to create binary masks that generate forged anomalous regions. However, the use of global random noise results in the generated anomalies appearing in image backgrounds. These forged anomaly samples may prompt the model to focus excessively on the background region, which results in the model being ineffective in detecting real anomalies, especially in industrial product scenes.

Towards this end, a novel Normal Image Guided Segmentation Framework (NIGSF) for anomaly detection is presented, as illustrated in Figure 1 (c). The proposed NIGSF is more perceptually consistent with human intuition for detecting abnormal images, *i.e.*, it utilizes normal samples as the reference to guide the segmentation model to identify “abnormal features” that deviate from the normal reference. This is called the normal image-guided anomaly detection paradigm. Our framework hinges on two main components: the Normal-Guided Network (NGN), and the Saliency Augmentation Module (SAM). Firstly, Normal-Guided Network (NGN) is proposed to extract valid normal features as reference features and guide the segmentation model to segment out abnormal regions. In NGN, we set up the contrast set, which is proposed as a candidate set for extracting guidance features for the input image. In addition, a normal feature extractor is developed to extract noiseless guidance features with detailed normal semantic information from the contrast set. Meanwhile, a simple yet effective multiscale guidance fusion is proposed to fully utilize the guidance features to guide the segmentation of anomalous features. The multiscale features account for the discrimination of tiny anomaly regions. Secondly, to further facilitate NIGSF learning to detect abnormal regions that deviate from normal patterns, a Saliency Augmentation Module (SAM) is proposed, which aims to generate forged anomalous samples that differ from the normal sample distribution to

participate in training. SAM introduces the saliency map and Perlin noise to forge anomalies in saliency regions rather than background regions while ensuring diversity and avoiding generating noisy samples that may hinder model learning. During training, NIGSF is expected to accurately segment the forged anomalies under the guidance of the normal image. Extensive experiments are conducted on three anomaly detection datasets. The performance of NIGSF over the state-of-the-art approaches demonstrates its superiority.

The main contributions of this paper can be summarized as follows:

- In this paper, the inadequacies of the current anomaly detection paradigm are analyzed, and a novel paradigm, anomaly detection guided by normal images, which aligns more closely with human intuition, is proposed.
- Based on the proposed anomaly detection paradigm, a novel Normal Image Guided Segmentation Framework (NIGSF) is proposed. It directly introduces normal image features as reference features and guides the segmentation model to identify “abnormal” classes that differ from the normal reference.
- A Saliency Augmentation Module (SAM) is proposed to generate diverse and valid forged anomaly samples, which facilitates the learning of NIGSF.

The remainder of the paper is organized as follows. Section II presents a brief overview of related work, including differences between recent studies and the proposed approach. Section III outlines the details of the proposed method. Section IV provides a detailed description of the experiments, containing the ablation experiments and the experimental results of the three data sets. Section V provides the conclusions.

II. RELATED WORK

Anomaly detection plays a pivotal role in visual tasks [20], [27]. The issue lies in the diversity of anomalous samples making it unfeasible to obtain representations of all anomalies, necessitating unsupervised methods which only require normal samples [29], [30], [31]. This challenge of detecting anomalies without prior knowledge of anomalous scenes is demanding. Initially, anomaly detection aimed to identify unlearned category samples, referred to as Out-of-Distribution (OOD) detection [32] or One-Class Classification [23]. Yet, in industrial and medical scenarios, locating anomalous regions in abnormal images becomes imperative [1], [20], [33], [34], thereby bifurcating unsupervised anomaly detection into two subtasks, namely, anomaly classification and anomaly segmentation. Various existing approaches include distribution-based, knowledge distillation-based, reconstruction-based, and forged anomalous sample-based methods.

Distribution-based methods [21], [22], [23] deduce normal sample distribution before identifying anomalies as samples falling outside these distributions. For instance, DeepSVDD [23] uses deep learning models to map the original image space to high-dimensional latent space. Normal samples' distribution is limited to certain regions in this space, enabling the identification of anomalies outside these regions.

These methods excel when normal and anomaly classes have a wide gap but falter with finer anomalies like leather scratches. Padim [24] further refines this approach by modeling the distribution of each normal image patch using multiple Gaussian mixture models [35], thus identifying outliers, yet precise anomaly segmentation remains elusive.

The knowledge distillation-based approach [2], [36] trains one or more student networks through a pre-trained teacher network and detects anomalies by the output's differences between the teacher and student networks for the samples. U-Std [2] utilizes multiple student networks and image patches as input to the network. However, it is inefficient in practice and can not achieve anomaly localization. MKDAD [36] uses only one student network and multi-scale feature distillation to encode normal samples closer together. It achieves anomaly localization relying on the gradients to locate anomalous regions. However, it is limited by the resolution of feature maps and does not enable accurate anomaly segmentation.

The reconstruction-based approaches usually utilize auto-encoder (AE) [37] to successfully reconstruct normal samples and poorly anomalous samples [19], [26], [38], [39]. Abnormal regions are detected and localized by comparing the difference between the original and reconstructed images. These methods are implemented directly through AE [19]. However, due to the strong reconstruction ability of AE, they can reconstruct anomalous samples well even if no anomalous samples are available in training. Some methods [4], [6], [7], [25], [40], [41], [42], [43] are proposed to reduce the reconstruction ability of AE by introducing memory networks. For example, MemAE [6] introduces a memory module after the encoder of the traditional AE, and PMB introduces a more efficient local memory module weakening anomaly recovery. However, the simple memory module makes the performance poor in the face of fine-grained anomalies. MNAD [7] further improves the original memory module and introduces a new loss function to improve the learning of the memory module. Unfortunately, these methods can still reconstruct unlearned objects well, and the anomaly segmentation performance of anomaly detection remains poor.

The forged anomalous sample-based approach is proposed to solve the problem of positive and negative sample imbalance in anomaly detection [20], [27]. After acquiring forged anomaly samples, supervised models can be used to solve the anomaly detection challenge. Cutpaste [27] generates forged abnormal samples that are added to the training set by using a part of the rectangular region of the original image to fill the forged region. However, the forged anomalies are not only rectangular in shape but also simple. Besides, using the gradient method to locate the anomalies, the pixel-level localization performance is poor. Inspired by the excellent performance of segmentation models in dense prediction tasks [44], [45], [46], [47], several studies [20], [48] have introduced segmentation networks in anomaly localization task. In [48], the forged images are first generated by the GAN as well as the semantic segmentation network, and then the distillation network is used to detect anomaly images. However, it is only applicable to anomaly classification and difficult to achieve anomaly localization. DRAEM [20] directly introduces segmentation

networks for segmenting forged anomalous samples. However, this is not effective due to the distributional gap between the generated and real anomalies. Towards this end, this paper proposes a more reasonable method to generate forged anomalous samples. Then, a novel anomaly detection framework is proposed, which utilizes normal features as guidance features to detect anomaly regions. The purpose of NIGSF is to learn and detect distributions that differ from the normal distribution, guided by the normal pattern.

III. THE PROPOSED METHOD

A. Overview

The overall architecture of the proposed Normal Image Guided Segmentation Framework (NIGSF) is shown in Figure 2, which contains a Normal Guided Network (NGN) and a Saliency Augmentation Module (SAM). The intuition of NIGSF is to identify unknown abnormal samples by comparing the differences with known normal samples which is similar to the detection of abnormal classes in humans. Towards this end, NGN (Sec. III-B) is proposed, which contains a normal feature extractor, the guidance feature fusion and a segmentation module. The purpose of the normal feature extractor is to extract the normal features with detailed information from only the normal samples. To facilitate feature-level guidance, this paper utilizes a simple yet effective guidance feature fusion method. Specifically, the features extracted from the input image (I or A) by the encoder of the segmentation module are multi-scale fused with the guidance features. The decoder of the segmentation module learns from the fused features and segments the “anomalies” that differ from the guidance features. To facilitate the segmentation module segmenting anomalies efficiently, SAM (Sec. III-C) is proposed. SAM aims to generate forged anomaly samples by available normal samples. It can generate reasonable forged anomaly samples in the saliency region rather than the background region. In the training phase, the generated forged anomalous samples serve as training samples along with normal samples to train the segmentation model to locate the forged regions. In the inference phase, the input samples are input to the segmentation module without SAM. In summary, NIGSF aims to enable the model to learn to identify content that differs from the guidance features as anomalies under normal sample guidance. Therefore, NIGSF differs from the traditional anomaly detection paradigm in that it is more in line with the anomaly detection principle, *i.e.*, identifies deviations from normal as abnormal.

Formally, A represents the input image and $y_A = \{y^{ij} | y^{ij} \in \{0\}\}_{i=1, j=1}^{N, N}$ is pixel-level label, where $N \times N$ represents the size of A , $y^{ij} = 0$ represents that the pixel is normal, $y^{ij} = 1$ represents that the pixel is abnormal. The contrast set is $C = \{C_1, C_2, \dots, C_n\}$, $y_C = \{y^{ij} | y^{ij} \in \{0\}\}_{i=1, j=1}^{N, N}$, where n denotes the number of samples in C . The sampled image is C_i as the input of the normal feature extractor. The auxiliary image is represented as B , $y_B = \{y^{ij} | y^{ij} \in \{1\}\}_{i=1, j=1}^{N, N}$. R represents the output of the normal feature extractor with the same size as C_i . The loss function of the normal feature extractor is $\mathcal{L}_N$. The image A is forged by the SAM to obtain

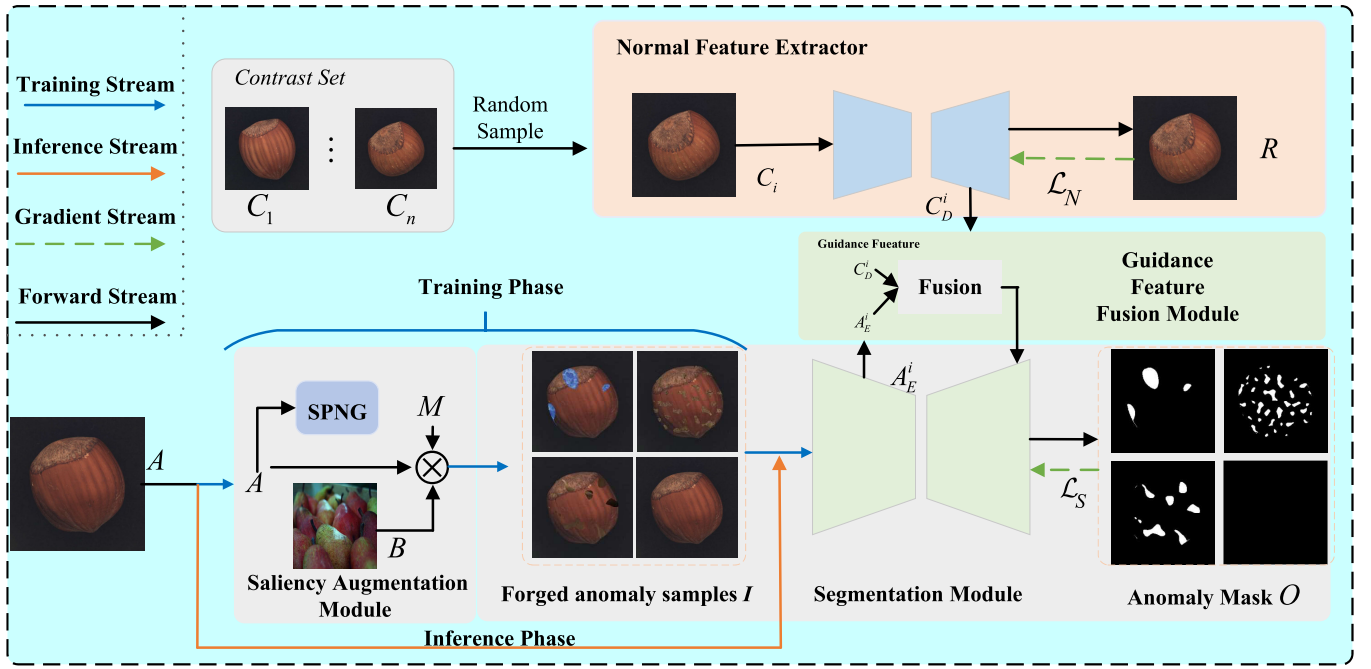


Fig. 2. Illustration of the proposed NISGF. A represents the input image. The contrast set C is first generated from A and available normal samples, then C_i is randomly sampled from C as the input of the normal feature extractor, which is responsible for extracting the noise-free and complete features as the guidance features that guide the segmentation module to segment anomaly regions. During training, the image A is input to the Saliency Augmentation Module (SAM) to generate the forged anomaly image I . The guidance features and features extracted by I are fed into the decoder of the segmentation network by the guidance feature fusion module. Finally, the decoder of the segmentation module identifies the difference between the input features and the guidance features segmenting the anomalous region to obtain the segmentation map O . The proposed NISGF utilizes $\mathcal{L}_N$ and $\mathcal{L}_S$ to train the model. SAM is discarded during the inference phase, and A is fed directly into the segmentation module to achieve anomaly detection.

the set of samples I , and $y_I = \{y^{ij} | y^{ij} \in \{0, 1\}\}_{i=1, j=1}^{N, N}$. O represents the output of the segmentation module as anomaly localization images. M represents the forged mask. The loss function of the segmentation network is $\mathcal{L}_S$.

B. Normal Guided Network

Since the high correlation among normal samples, it is natural to consider normal sample comparisons for identifying anomalies. In this regard, we first introduce the concept of a contrast set whose images serve as reference samples for the input samples. We then extract features from the reference images using a normal feature extractor as guidance features. Finally, the anomaly detection results are obtained by guidance feature fusion and the segmentation module.

1) *Contrast Set*: Given the substantial intra-class variations exhibited by samples, even normal ones, low-level attributes such as texture and color tend to differ. Consequently, randomly selecting normal samples as reference images poses a challenge in ensuring the extracted features suitable for guiding the detection of anomalous samples. This paper introduces the contrast set. The contrast set is composed of normal samples that closely resemble input image A , allowing for the controlled similarity of low-level information and high-level semantic information between the guidance features and features extracted from A . As a result, the segmentation module is better guided to locate tiny anomalies using the guidance features obtained from the segmentation module.

Specifically, the proposed approach utilizes a feature extractor, where the output $q \in \mathbb{R}^{512 \times 8 \times 8}$ of the extractor's last layer

serves as the features for measuring similarity. The output q is then flattened into a vector $p_A \in \mathbb{R}^{512}$ by the mean-pooling layer. p_A obtained from image A is used to calculate the cosine similarity T with the training normal images p_{Train} .

$$T = \{T^1, T^2, \dots, T^j, \dots, T^m | T^j = \frac{p_A \cdot p_{Train}^j}{\|p_A\| \times \|p_{Train}^j\|}\}, \quad (1)$$

where m is the number of available normal samples to construct the contrast set. Then, T is arranged as T' in descending order of similarity. The top n in T' that correspond to the original images are used to constitute the contrast set of A . It is important to note that the contrast set of image A does not comprise A itself since this can prevent the network from excessively attending to low-level features (see Sec. IV-E). During the training phase, images are randomly sampled from the contrast set and used as input for the normal feature extractor. On the other hand, in the reference phase, only normal samples that are most similar to the input samples A are included in the contrast set (Table VIII and ablation experiments (5) give more explanation.).

2) *Normal Feature Extractor*: To guarantee that the extracted features comprise comprehensive and detailed information, a normal feature extractor is employed, as shown in Figure 3. In particular, C_i , sampled randomly from the contrast set, is inputted into the extractor to ensure that the extracted guidance features possess normal semantic information. Specifically, the normal feature extractor with a standard

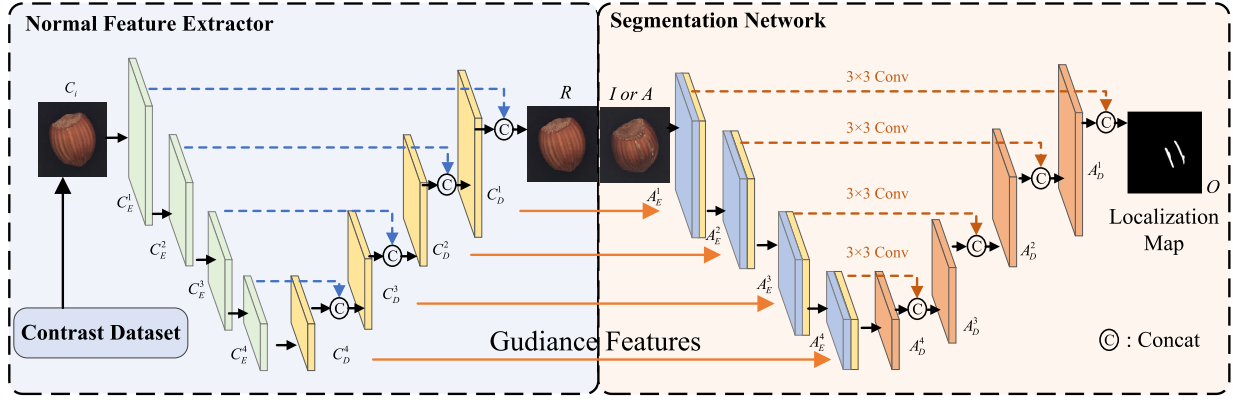


Fig. 3. Illustration of the guidance feature fusion module and segmentation module. C_i is randomly sampled from the contrast set C and used as the input of the normal feature extractor. The features of C_i are extracted by the normal feature extractor as the guidance features $C_D = \{C_D^i\}_{i=1}^4$. The segmentation module extracts the features $A_E = \{A_E^i\}_{i=1}^4$ of the input image I or A (During the inference phase, A is input directly into the segmentation module without SAM). The C_D as guidance feature to provide feature-level guidance. The fused results are subjected to a standard 3×3 convolution layer to extract discriminative features, which utilize skip connections to guide the decoder to obtain anomalous localization map O .

AE composition is trained by a reconstruction task. The reconstruction task compresses and then recovers the image features to ensure that the output of the decoder is normal features. In addition, the multi-scale ensures the extraction of high-dimensional semantic and low-dimensional information. Therefore, the multi-scale features output from the normal feature extractor decoder are used as guidance features.

3) *Guidance Feature Fusion*: To achieve normal features as the guidance for input images, we adopt a simple yet effective guidance feature fusion, as shown in Figure 3. We utilize guidance features and the encoder of segmentation module to achieve fusion. In each step, features with the same resolution are fused through each layer of encoder to achieve feature level fusion. Formally, the multi-scale features extracted by a normal feature extractor network are $C_E = \{C_E^i\}_{i=1}^4$. The output of the decoder is $C_D = \{C_D^i\}_{i=1}^4$, where C_D is defined as the guidance features. The output of the encoder of the segmentation network is $A_E = \{A_E^i\}_{i=1}^4$ and the output of the decoder is $A_D = \{A_D^i\}_{i=1}^4$. The proposed approach directly fuses C_D and A_E , which ensures the segmentation of tiny anomalous regions. It can be expressed as:

$$AC_i = \text{cat}(C_D^i, A_E^i), \quad i = 1, 2, 3, 4, \quad (2)$$

where $\text{cat}(\cdot)$ represents two features concatenated along the channel dimension. Then, we further process AC_i by 3×3 convolutional layer to implement feature-level guidance.

$$\widehat{AC}_i = \sigma(BN_i(\text{Conv}_i(AC_i))), \quad (3)$$

where $BN_i(\cdot)$ denotes batch normalization layer, $\sigma(\cdot)$ denotes the ReLU activation function. AC_i and $\widehat{AC}_i$ are fed to the next encoder layer and to the decoder after the skip connection, respectively.

4) *Segmentation Module*: The segmentation module consists of an autoencoder. The encoder is responsible for extracting the multi-scale features of the input image. The fused features are input to the decoder, then gradually obtains the segmentation result by transposition convolution. Taking layer i of the decoder as an example. First, its input is the

output A_D^{i-1} of layer $i - 1$.

$$\widehat{A}_D^i = \text{TrConv}_i(A_D^{i-1}), \quad (4)$$

where $\text{TrConv}_i(\cdot)$ is $2 \times$ transposed convolution, $\widehat{A}_D^i$ is output of the transposed convolution. Meanwhile, the guidance features C_D^i are fused with A_E^i to obtain a new tensor $AC_i \in \mathbb{R}^{h \times w \times 2c}$. We utilize the fused features to generate further $\widehat{AC}_i$ and then obtain the decoder output A_D^i at layer i . A_D^i is represented as follows:

$$A_D^i = \sigma(BN_i^2(\text{Conv}_i^2(\sigma(BN_i^1(\text{Conv}_i^1(\text{cat}(\widehat{AC}_i, \widehat{A}_D^{i-1})))))). \quad (5)$$

where $BN_i^1(\cdot)$ and $BN_i^2(\cdot)$ denote two batch normalization layers, Conv_i^1 and Conv_i^2 represent 3×3 with stride = 1. By performing the aforementioned operations iteratively, a segmentation map O is obtained. Each pixel value of the localization image O indicates the likelihood of each pixel being discriminated as anomalous. Following [20], O is processed through the mean-pooling layer to aggregate local anomaly information. Furthermore, the proposed method utilizes the maximum pixel value of O rather than the average value as the anomaly score of the sample to eliminate any adverse effects caused by noise.

C. Saliency Augmentation Module

SAM aims to generate reasonable forged anomalous samples that are close to the real anomaly distribution and suitable for model learning, which includes a Saliency Perlin Noise Generator (SPNG) and a Forged Sample Generator (FSG), as shown in Figure 4. We follow DRAEM [20] to introduce random Perlin noise. However, the mask generated by DRAEM using global noise may cause the model to focus excessively on the background region, which is detrimental to industrial anomaly detection. SAM further introduces saliency information. SPNG generates saliency Perlin noise to make the generated anomaly masks lie in the saliency region instead of the background region. Meanwhile, to generate diverse

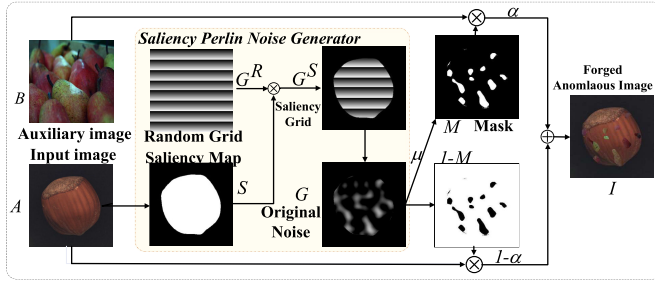


Fig. 4. Illustration of the **Saliency Augmentation Module (SAM)**. The input of the SAM consists of three parts: the original input image A , the auxiliary image B and the saliency map S . The Saliency Perlin Generator generates the saliency Perlin noise G . The mask M is obtained from the saliency Perlin noise G using an adaptive threshold segmentation strategy. Finally, the original image A , auxiliary image B and mask (M and $1 - M$) are processed to generate the forged anomaly image I . G^R represents the random grid.

masks, an adaptive threshold scheme is proposed to binarize the saliency Perlin noise. Then, FSG utilizes the auxiliary samples, the mask generated by SPNG and the original image to generate the forged anomaly samples.

Specifically, as shown in Figure 4, the input of SAM is the original image A , the saliency map S extracted using the saliency method [49], and the auxiliary image B . First, a random grid G^R is generated with the same size as the original image A . Second, the saliency grid G^S is obtained by Hadamard product between saliency map S and the original grid G^R , which is used to generate the original noise G in the saliency region. Third, by setting the threshold μ , the foreground mask (M) and the background mask ($1 - M$) are segmented. In this paper, an adaptive threshold scheme is developed associated with the saliency map as follows:

$$\mu = a + \frac{\text{sum}(S)}{N * N} \times b, \quad (6)$$

where $[a, a + b]$ denotes the value domain of μ , $\text{sum}(S)$ denotes the sum of the values of S , and N represents the size of image A . Finally, FSG fuses the image A , the auxiliary image B , the background mask ($1 - M$) and the foreground mask (M) to generate a forged anomaly sample I . The specific methods are as follows:

$$I = A \odot (1 - M) + \alpha \times (B \odot M) + (1 - \alpha) \times (A \odot M), \quad (7)$$

where α denotes the proximity of the simulation to the normal sample distribution, $\odot$ denotes the Hadamard product. The label of I is composed of the labels of A , B , and M , i.e., $y_I = M \odot y_B + (1 - M) \odot y_A$. To ensure the diversity of the generated forged anomaly samples, NIGSF randomly samples auxiliary image B from a large-scale dataset and applies data augmentation operations when generating each forged anomaly sample.

However, existing saliency detection methods identify the whole image as a background when targeting textured images. Each pixel value on the saliency map is 0, which is obviously not reasonable. Therefore, we used the following way to

generate G^S . Formally,

$$G^S = \begin{cases} G^R, & \text{sum}(S) = 0, \\ G^S, & \text{otherwise,} \end{cases} \quad (8)$$

D. Loss Function

The proposed approach contains losses $\mathcal{L}_N$ and $\mathcal{L}_S$. For the normal feature extractor, the l_2 loss is introduced. Besides, the $SSIM$ loss function [50] is also introduced to enhance the reconstruction result so that the output guidance features are completely noise-free normal features. Following [19], $SSIM$ loss is obtained from Equation (10). It is expressed as follows:

$$\mathcal{L}_{l_2} = \|R - C\|_2, \quad (9)$$

$$\mathcal{L}_{SSIM} = \frac{1}{N \times N} \sum_{i=1}^N \sum_{j=1}^N 1 - SSIM_{(i,j)}^k(R, C), \quad (10)$$

where R and C are the reconstruction image and input image of normal feature extractor. $SSIM_{(i,j)}^k(R, C)$ denotes the structural similarity index of a block of size $k \times k$ centred at (i, j) . For the segmentation module, the Focal loss [51] is introduced to alleviate the imbalance between normal and anomalous pixels. $\mathcal{L}_S$ is expressed as follows:

$$\mathcal{L}_S = \begin{cases} -(1 - p)^\tau \log(p), & y_I^{ij} = 1, \\ -p^\tau \log(1 - p), & y_I^{ij} = 0. \end{cases} \quad (11)$$

where p represents the probability of pixel (i, j) prediction. Finally, the total loss of the proposed NIGSF is expressed as:

$$\mathcal{L} = \lambda \times \mathcal{L}_N + \mathcal{L}_S = \lambda \times (\mathcal{L}_{l_2} + \mathcal{L}_{SSIM}) + \mathcal{L}_S, \quad (12)$$

where λ is set to measure the importance of the two loss functions.

IV. EXPERIMENTS

In this section, we introduce the experimental datasets, the experimental setup, and anomaly classification and localization performance. In addition, we conduct extensive ablation experiments to demonstrate the effectiveness of each proposed module.

A. Datasets

1) *MVTec*: MVTec [1] is a challenging anomaly detection dataset. It contains a series of industrial real abnormal and normal images. The MVTec dataset includes 15 different categories, containing both texture images (e.g., “leather”) and non-textured images (e.g., “cable”). Anomaly classification and anomaly localization can be evaluated. In the training phase only normal samples for training and in the reference phase different anomalies need to be detected. All images are high-resolution images, and some anomalous regions are tiny, such as “broken screw”.

2) *KolektorSDD2*: KolektorSDD2 [55] is a surface anomaly detection dataset from real scenes. The dataset consists of color images of defective production items captured by a vision inspection system. The images in the dataset are captured in a controlled environment and are similar in size, about 230×630 . The dataset is complex and accurately annotated. In the original dataset, the training set contains a large number of abnormal samples. In this work, we first process the original dataset to contain only normal samples for training. Finally,

TABLE I

COMPARISON OF THE PROPOSED METHOD AND UNSUPERVISED ANOMALY DETECTION METHODS ON MVTec DATASET WITH AUROC (%). THE BEST RESULTS ARE MARKED IN BOLD. BOTH DRAEM₁ AND DRAEM₂ ARE REPRODUCED FROM THIS PAPER, WHERE DRAEM₁ UTILIZES THE U-NET MODEL WITH THE LARGER PARAMETERS FROM [20] AND DRAEM₂ USES THE SAME UNET MODEL AS NIGSF

Methods	Carpet	Screw	Cable	Wood	Leather	Tile	Grid	Toothbrush	Zipper	Bottle	Hazelnut	Capsule	Metal nut	Pill	Transistor	MEAN
GANomaly [52]	82.1	100	71.1	92.0	80.8	72.0	74.3	70.0	74.4	79.4	87.4	72.1	69.4	67.1	80.8	78.2
ARAFD [26]	70.6	100	83.2	92.3	86.2	73.5	88.3	100	87.6	94.1	85.5	68.1	66.7	78.6	84.3	83.9
MKDAD [36]	79.3	83.3	89.2	94.3	95.1	91.6	78.0	92.2	93.2	99.4	98.4	80.5	73.6	82.7	85.6	87.7
U-Std [2]	91.6	54.9	86.2	97.7	88.2	99.1	81.0	95.3	91.9	99.0	93.1	86.1	82.0	87.9	81.8	87.7
DAAD+ [25]	86.6	98.7	84.4	98.2	86.2	88.2	95.7	99.2	85.9	97.6	92.1	76.7	75.8	90.0	87.6	89.5
RIAD [53]	84.2	84.5	81.9	93.0	100	98.7	99.6	100	98.1	99.9	83.3	88.4	88.5	83.8	90.9	91.7
MAD [54]	95.5	81.2	94.0	97.6	100	97.4	92.9	95.8	97.9	100	98.7	92.3	93.1	83.4	95.9	94.4
Cutpaste [27]	93.1	86.3	80.6	98.6	100	93.4	99.9	98.3	99.4	98.3	97.3	96.2	99.3	92.4	95.5	95.2
Padim [24]	99.8	85.8	92.7	99.2	100	98.1	96.7	96.1	90.3	99.9	92.0	91.3	98.7	93.3	97.4	95.5
UniAD [3]	99.9	91.9	97.6	97.9	100	99.0	98.5	95.0	96.7	100	99.9	85.3	99.0	88.3	100	96.6
DRAEM ₁ [20]	97.0	93.9	91.8	99.1	100	99.6	99.9	100	100	99.2	100	98.5	98.7	98.9	93.1	97.9
DRAEM ₂ [20]	86.4	79.2	93.5	97.4	99.7	100	99.8	97.5	99.5	96.6	98.8	93.2	87.1	91.3	92.6	94.2
NIGSF	96.5	86.6	85.6	99.7	100	100	100	100	99.6	96.0	99.8	93.9	99.7	97.2	95.3	96.7

the number of training samples is 2085, and the test samples is 1004. The anomalies include scratches or large surface defects with different shapes, sizes, and colors.

3) *DAGM*: DAGM [61] is a well-known dataset for surface defect detection. It contains ten categories of computer-generated gray images of surfaces and various defects. Early, only six categories of data were publicly evaluated, and subsequently, four categories were publicly available. The anomaly label of this dataset is inaccurate. Therefore, only the anomaly classification performance of the DAGM dataset is evaluated.

B. Baselines and Evaluation Metric

1) *Baselines*: NIGSF is compared with state-of-the-art deep methods including fully supervised methods, i.e., CCNN [56] and ENSDD [57], weakly supervised methods, i.e., CADN-W18 [58], CADN-W18 (KD) [58], and MSDD [55], as well as unsupervised methods, i.e., GANomaly [52], MKDAD [36], U-Std [2], DAAD+ [25], RIAD [53], MAD [54], Cutpaste [27], Padim [24], RD [62], UniAD [3], and SPADE [63]. Fully supervised methods use all the normal samples with pixel-level labels. The weakly supervised method uses image-level labelled abnormal samples with all normal samples or a part of pixel-level labelled abnormal samples with all normal samples. The unsupervised method only uses normal samples. The MKDAD, U-Std, RD, UniAD, and Cutpaste use the pre-trained model. The Cutpaste and DRAEM use forged anomalous samples.

2) *Evaluation Metric*: Following previous work [1], [36], [64], image-level and pixel-level Area Under the Receiver Operating Characteristic Curve (AUROC) are introduced to evaluate the performance of anomaly classification and anomaly localization. The metric of AUROC does not reflect the true performance in evaluating the localization performance. However, the percentage of anomalous pixels is small in the test images, which makes the overall false positive rate perform at a low-level [20]. Therefore, the pixel-level average precision (AP) metric is introduced. It evaluates the accuracy of the overall pixel discrimination and the localization performance together with the pixel-level AUROC. $AUROC_{det}$,

$AUROC_{loc}$ and AP_{loc} represent AUROC for anomaly image classification, AUROC for anomaly image localization, and AP for anomaly image localization, respectively. Following [55], on the DAGM dataset, the image level AUROC, AP, F1, TPR, TNR, ACC, and $mACC = (TPR + TNR)/2$ are introduced to evaluate the performance of anomaly classification.

C. Implementation Details

1) *Sets*: The number of contrast set n is set to 5. The auxiliary images in SAM are randomly sampled from the DTD [65]. We follow the image augmentation method in the DRAEM and adjust the ratio of forged abnormal images to normal images to 3:1 during the training phase.

2) *Experimental Setting*: The proposed model is trained on a single RTX TiTAN GPU using PyTorch. It is trained with a mini-batch size of 16 using the Adam optimizer. The learning rate is set to 1×10^{-4} . The learning rate decays to 0.1 times the current one at $800 \times [0.5, 0.7, 0.9]$ iterations, respectively. The hyperparameter a is set to 0.4, b is set to 0.2 in Eq. (6), and the input image size is set to 256×256 . In Eq. (7) $a \in [0.1, 1]$. In Eq. (12) λ is set to 1 and in Eq. (11) τ is set to 2.

D. Comparison With State-of-the-Art Methods

1) *Anomaly Image Classification*: For the anomaly classification task, we evaluate the performance of the proposed method on MVTec, DAGM, and KolektorSDD2 datasets.

Table I presents a comparison between the anomaly classification performance of NIGSF and unsupervised anomaly detection methods on MVTec dataset. The results indicate that NIGSF is a competitive method, achieving a mean $AUROC_{det}$ of 96.7%. Specifically, NIGSF outperforms the remaining unsupervised methods in four categories, such as “leather” and “Toothbrush” categories, achieving 100% in terms of $AUROC_{det}$. Additionally, it demonstrates the best performance in eight categories of MVTec, which verifies the effectiveness of introducing normal sample features as reference features. In comparison with the recently advanced UniAD method that incorporates Transformer and pretrained

TABLE II

COMPARISON OF THE PROPOSED METHOD AND UNSUPERVISED ANOMALY DETECTION METHODS ON DAGM DATASET WITH AUROC (%). N IN MSDD INDICATES THAT MSDD USES N PIXEL-LEVEL LABELLED ANOMALY SAMPLES. THE BEST RESULTS ARE MARKED IN BOLD

Type	Methods	10 categories						6 categories	
		AUROC	AP	F1	TPR	TNR	CA	CA	mACC
Supervised	CCNN [56]	99.6	-	-	99.9	99.4	99.6	99.2	99.4
	ENSDD [57]	-	99.0	-	99.4	99.9	-	-	99.8
Weakly-supervised	CADN-W18 [58]	-	-	63.2	-	-	86.2	-	-
	CADN-W18(KD) [58]	-	-	65.8	-	-	87.6	-	-
supervised	MSDD(N=0) [55]	74.0	86.1	74.6	-	-	89.7	85.4	81.4
	MSDD(N=5) [55]	91.5	94.9	92.3	-	-	92.9	88.1	91.6
Unsupervised	F-AnoGAN [59]	57.5	19.5	27.8	-	-	79.7	81.7	54.6
	U-std [2]	86.4	66.8	67.1	-	-	84.3	79.3	78.5
	Staaer [60]	83.0	-	-	-	-	-	-	-
	RIAD [53]	78.6	-	-	79.2	69.1	70.4	-	-
	MAD [54]	82.4	-	-	78.7	85.7	66.2	-	-
	PaDim [24]	95.0	-	-	83.3	97.5	95.7	-	-
	DRAEM [20]	99.0	-	-	96.5	99.4	98.5	-	-
	NIGSF	99.9	99.7	98.2	98.1	99.8	99.5	99.6	99.1

models, NIGSF still remains more competitive. In addition, we compare it with the current state-of-the-art DRAEM method. We reproduce the results of DRAEM, denoted as DRAEM₁. Furthermore, we find that DRAEM uses a larger Unet model, so we replace this model with the same Unet model as the proposed NIGSF method and perform anomaly detection in MVTec-AD. The results are denoted as DRAEM₂. It can be found that DRAEM relies on a specific UNet model. NIGSF significantly outperforms DRAEM when the same network structure is used. The above experimental results show the effectiveness of NIGSF for anomaly classification on MVTec dataset.

In Table II, we compare the anomaly classification performance of NIGSF with unsupervised, weakly, and supervised anomaly detection methods on DAGM dataset. The experimental results indicate that NIGSF outperforms the unsupervised and weakly supervised methods in 10 and 6 categories, respectively, highlighting its superiority over these methods. The proposed method achieves AUROC 99.9%, indicating that NIGSF can solve the anomaly classification challenge of grayscale images well. Specifically, the anomaly classification performance of NIGSF without real anomaly supervision information is the same as the supervised CCNN and ENSDD. Importantly, there is a large distribution gap between the forged anomalous samples and the grayscale anomalies of DAGM, yet the anomalous samples in DAGM can be detected. This is due to the fact that NIGSF by introducing normal samples as guidance, the model learns to detect different from guidance images as anomalies. NIGSF generalizes well for grayscale image anomaly detection.

In Table III, we compare the anomaly classification performance of NIGSF and unsupervised and weakly supervised anomaly detection methods on KolektorSDD2 dataset. The experimental results show that NIGSF achieves the best performance for the weakly supervised methods and unsupervised methods. NIGSF outperforms unsupervised methods U-std 21% and F-AnoGan 30% and outperforms weakly supervised

TABLE III

COMPARISON OF NIGSF AND UNSUPERVISED ANOMALY DETECTION METHODS ON KOLEKTORSDD2 DATASET WITH AP_{det} (%). THE BEST RESULTS ARE MARKED IN BOLD

Type	Methods	AP_{det}
Weakly-supervised	MSDD(N=0) [55]	73.3
	MSDD(N=16) [55]	83.2
Unsupervised	F-AnoGan [59]	55.0
	U-Std [2]	65.3
	NIGSF	86.3

method MSDD 3%, indicating the effectiveness of introducing normal features to guide the segmentation module. Even though real anomaly samples are lacking, the proposed normal image guided anomaly detection paradigm can effectively solve the anomaly detection challenges of multiple scenes.

2) *Anomaly Localization*: To evaluate the performance of NIGSF in anomaly localization task, we conduct experiment on the MVTec dataset. The experimental results of the proposed method and unsupervised anomaly detection methods are shown in Table IV. The proposed NIGSF achieves mean $AUROC_{loc}$ 97.2% and mean AP_{loc} 73.8%. The proposed method achieved similar results with PaDim and DRAEM in $AUROC_{loc}$. However, NIGSF is 18% higher than PaDim and 5% higher than DRAEM in AP_{loc} . For the “toothbrush” category, the proposed NIGSF is the only one that achieves 78.9% in the AP_{loc} metric, exceeding the rest of the methods by at least 24%, validating the efficiency of the segmentation paradigm using normal image guidance. The remaining unsupervised anomaly localization methods have similar findings while obtaining good localization performance for anomalous regions is difficult. Since the segmentation module of NIGSF has prior knowledge of normal patterns, it can detect all patterns different from normal guidance as anomalies, which

TABLE IV
ANOMALY LOCALIZATION PERFORMANCE OF THE PROPOSED METHOD AND OTHER ANOMALY DETECTION METHODS ON THE MVTEC DATASET

Metric	METHODS	Carpet	Screw	Cable	Wood	Leather	Tile	Grid	Toothbrush	Zipper	Bottle	Hazelnut	Capsule	Metal_nut	Pill	Transistor	MEAN
$AUROC_{loc}$	U-Std [2]	93.5	97.4	91.9	92.1	97.8	92.5	89.9	97.9	95.6	97.8	98.2	96.8	97.2	96.5	73.7	93.9
	RIAD [53]	96.3	98.8	84.2	85.8	99.4	89.1	98.8	98.9	97.8	98.4	96.1	92.8	92.5	95.7	87.7	94.2
	Cutpaste [27]	98.3	96.7	90.0	95.5	99.5	90.5	97.5	98.1	99.3	97.6	97.3	97.4	93.1	95.7	93.0	96.0
	SPADE [63]	97.5	98.9	97.2	88.5	97.6	87.4	93.7	97.9	96.5	98.4	99.1	99.0	98.1	96.5	94.1	96.5
	PaDim [24]	99.0	94.4	96.7	94.1	99.0	94.1	98.6	98.8	98.4	98.2	98.1	98.6	97.3	95.7	97.6	97.4
	DRAEM [20]	95.5	97.6	94.7	96.4	98.6	99.2	99.7	98.1	98.8	99.1	99.7	94.3	99.5	97.6	90.9	97.3
	UniAD [3]	98.0	97.4	96.8	93.4	98.3	91.8	94.6	97.8	96.0	98.1	98.8	97.9	95.7	95.1	98.7	96.6
	NIGSF	97.4	97.2	93.3	97.3	98.9	99.7	99.2	99.6	98.2	96.3	99.5	97.4	99.5	99.5	84.7	97.2
AP_{loc}	U-Std [2]	52.2	7.8	48.2	53.3	40.9	65.3	10.1	37.7	36.1	74.2	57.8	25.9	83.5	62.0	27.1	45.5
	RIAD [53]	61.4	43.9	24.4	38.2	49.1	52.6	36.4	50.6	63.4	76.4	33.8	38.2	64.3	51.6	39.2	48.2
	PaDim [24]	60.7	21.7	45.4	46.3	53.5	52.4	35.7	54.7	58.2	77.3	61.1	46.7	77.4	61.2	72.0	55.0
	DRAEM ₁ [20]	53.5	58.2	52.4	77.7	75.3	92.3	65.7	44.7	81.5	86.5	92.9	49.4	96.3	48.5	50.7	68.4
	RD [62]	64.1	54.8	59.2	53.6	52.4	50.4	47.6	54.5	60.6	79.4	64.5	45.8	80.9	80.0	55.7	60.2
	NIGSF	70.5	43.8	62.0	84.4	73.2	97.2	66.9	78.9	83.2	81.5	83.7	51.6	96.8	93.2	40.1	73.8

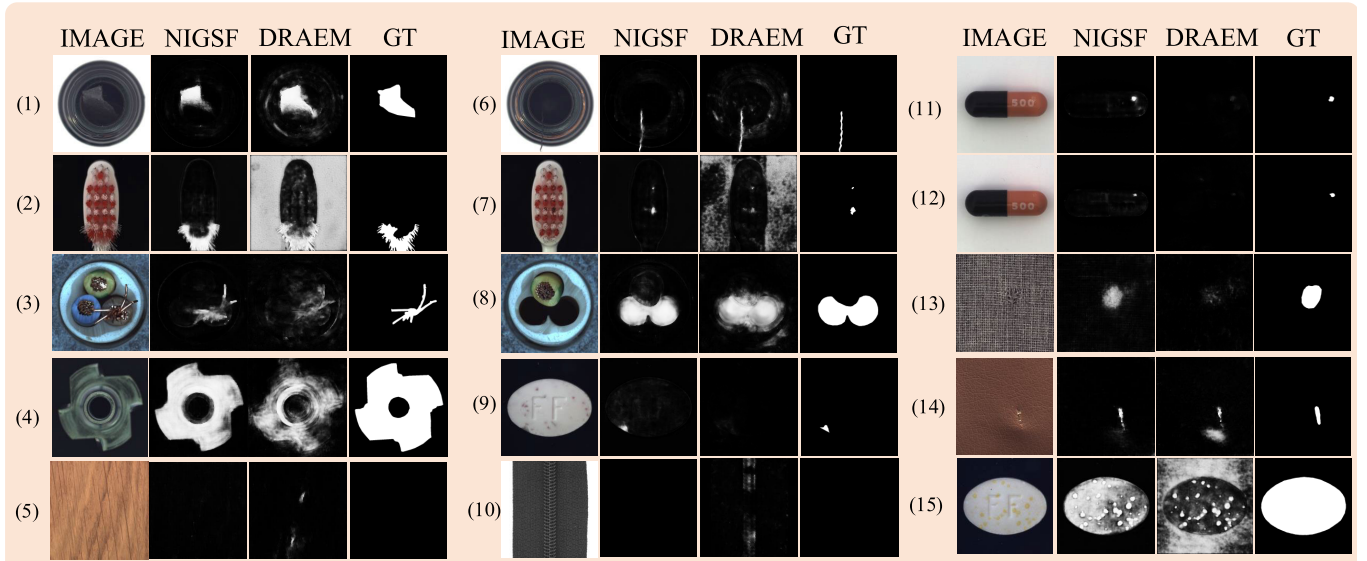


Fig. 5. Anomaly segmentation results of the proposed NIGSF with DRAEM [20], where (5) and (10) are normal samples and the rest are abnormal samples. “IMAGE” denotes the test image. “NIGSF” denotes the results of the proposed method. “DRAEM” denotes the results of DRAEM. “GT” represents groundtruth. The accuracy of the proposed NIGSF segmentation significantly exceeds the current state-of-the-art methods.

is not available in other methods. In summary, the proposed NIGSF method achieves state-of-the-art localization results.

In Figure 5, we visualize the anomaly localization results of the proposed method and the DRAEM. It can be seen that NIGSF has the following advantages. First, NIGSF is more accurate for detection of normal regions in the abnormal samples, as shown in (1), (5), (6), and (10). The localization map of DRAEM has many mislocalized anomalous regions. NIGSF directly utilizes normal sample features as guidance features, thus ensuring the validity of the guidance information. Therefore, the segmentation module can correctly identify the normal features and segment the abnormal ones that differ from the normal guidance features. Second, NIGSF performs better for detail boundary localization, as shown in (3) and (8). It demonstrates the advantage of using multi-scale features for guidance in NIGSF. The low-level features

effectively provide guidance about the boundary information, resulting in excellent localization performance of the segmentation module. Third, (9), (11), and (12) show that our method has advantages in localizing tiny anomalous regions. The detailed feature information acts as guidance features that allow the model to effectively locate small anomalies. Fourth, (2), (7), and (15) show that DRAEM is wrong in predicting the background. NIGSF addresses this problem in the Saliency Augmentation module. SAM introduces saliency map that makes the model more focused on foreground content. However, Figure 7 (C) demonstrates that NIGSF can also precisely localize to anomalies outside the saliency region. Finally, NIGSF is more sensitive for locating anomalous regions. As shown in (13), the anomalies are subtle textural differences. NIGSF can still accurately locate detailed anomalies, which are unavailable with DRAEM.

TABLE V

ABLATION STUDIES ON NGN. “- / - / -” REPRESENTS THE AVERAGE RESULT OF 5 TEXTURE CLASSES, THE AVERAGE RESULT OF 10 OBJECT CLASSES, AND THE OVERALL AVERAGE RESULT, RESPECTIVELY

	NGN	AUROC _{det}	AUROC _{loc}	AP _{loc}
baseline	$\times$	98.7 / 92.7 / 94.7	98.4 / 92.2 / 94.3	74.1 / 47.9 / 56.7
baseline+NGN	Denoising	99.2 / 92.5 / 94.8	98.5 / 92.0 / 94.1	78.7 / 60.2 / 66.3
baseline+NGN	Reconstruction	99.2 / 95.4 / 96.6	98.5 / 96.5 / 97.2	78.4 / 71.5 / 73.8

TABLE VI
ABLATION STUDIES ON SAM

		(1)	(2)	(3)	(4)
NGN	Reconstruction task	$\times$	$\times$	$\checkmark$	$\checkmark$
SAM	SPNG	$\times$	$\checkmark$	$\times$	$\checkmark$
	PNG	$\checkmark$	$\times$	$\checkmark$	$\times$
Metric	AUROC _{det}	93.8	94.7 ($\uparrow$ 0.9)	95.1	96.7 ($\uparrow$ 1.5)
	AUROC _{loc}	93.5	94.3 ($\uparrow$ 0.8)	95.0	97.2 ($\uparrow$ 2.2)
	AP _{loc}	54.3	56.7 ($\uparrow$ 1.4)	67.7	73.8 ($\uparrow$ 5.1)

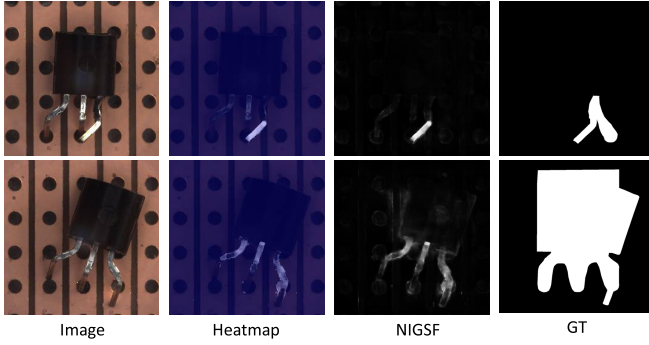


Fig. 6. Illustration of the localization map and Ground Truth of some samples in the “Transistor” category. The anomaly detection results of these images need to predict not only the regions where anomalies appear, but also the regions where components are missing.

a) *Failure cases*: Although NIGSF achieves good results in most categories, it does not perform well in the “transistor” category. Figure 6 shows the localization results of the samples in “transistor” category. Groundtruth of some abnormal samples is marked not only with “component bending” but also with semantically missing regions. The model is required to further understand the fine-grained semantics of normal samples: the position of the component is related to whether the sample is abnormal or normal. However, NIGSF only locates the anomalous component but fails to predict the original component location.

E. Ablation Study

1) *Normal Image Guided Anomaly Detection Paradigm*: To validate the effectiveness of the proposed paradigm, ablation experiments were conducted without normal image guidance. In other words, the segmentation module directly segments forged anomalous and normal regions without guidance features. The results are shown in Table V. It can be clearly

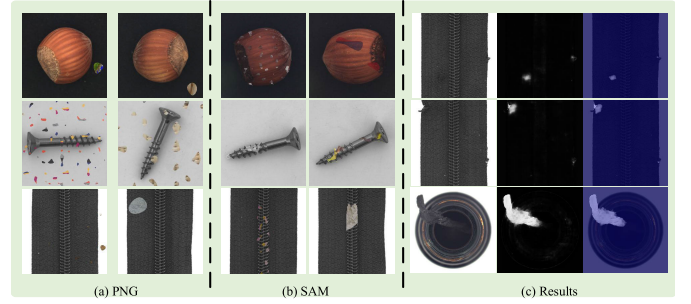


Fig. 7. The forged anomalous samples generated by DRAEM (a) and SAM (b) are illustrated. (c) shows that our method can locate the anomalous regions beyond the saliency regions.

found that the anomaly detection performance decreases with the lack of normal image guidance. With image guidance, both image anomaly detection and localization are significantly improved. Particularly, AP_{loc} decreases by 17% compared to the method that introduces normal guidance. Even though the SAM can generate forged anomalous samples close to the real anomalies to participate in training, there is still a distribution gap with the real anomalous samples. The normal feature thus plays a key role in guiding the segmentation model by explicitly providing normal distribution information that contributes to detecting abnormal distribution. The experimental results validate the effectiveness of the normal image guided anomaly detection paradigm.

2) *Saliency Perlin Noise Generator*: To validate the effectiveness of the proposed Saliency Perlin Noise Generator (SPNG), experiments were conducted to explore the differences between it and traditional generation methods. Figure 7 (A) shows the forged anomalous samples generated by DRAEM, while Figure 7 (B) shows the forged anomalous samples generated by the proposed Saliency Perlin Noise Generator (SPNG). Obviously, NIGSF focuses on saliency regions

TABLE VII

THE IMPACT OF MODEL SIZE ON THE PERFORMANCE OF THE METHODS. THE DRAEM₁, DRAEM₂ USE THE ORIGINAL STRUCTURE AND THE NETWORK STRUCTURE OF THIS PAPER, RESPECTIVELY

Method	Network From		Parameter		Metric		
	DRAEM	NIGSF	PNG	SPNG		AUROC _{det}	AP _{loc}
DRAEM ₁	✓	✗	✓	✗	97M	97.9	68.3
DRAEM ₂	✗	✓	✓	✗	70M	94.2	55.4
NIGSF	✗	✓	✓	✗	70M	94.8 (↑ 0.6)	66.3 (↑ 10.9)
NIGSF	✗	✓	✗	✓	70M	96.6 (↑ 2.4)	73.8 (↑ 18.4)

to generate anomalies, while DRAEM tends to generate more noisy samples in the background, especially in the category of small foreground objects (e.g., “screw”). Table VI shows the results of the ablation experiments in two settings. Comparing (1) and (2), (3) and (4), the anomaly localization and anomaly classification results of NIGSF using PNG are lower than those using SPNG, which validates the effectiveness of SPNG. SPNG reduces the noise sample generation and improves the learning efficiency of effective samples. Specifically, although NIGSF produces anomalies only in the saliency region, however, we can still obtain good localization results on regions outside the saliency region, such as the zipper shown in Figure 7 (C) (the saliency map contains only the “zipper” not the “cloth”, but anomalies on the “cloth” can be detected).

3) *Reconstruction Task and Denoising Task*: To validate the influence of self-supervised tasks in the normal feature extractor, different self-supervised tasks are selected to validate the performance. Different self-supervised tasks have advantages and disadvantages for NIGSF, and we examined the reconstruction task and denoising task in our experiments. Specifically, we choose the reconstruction task and denoising task as the training method for the normal feature extractor to verify their effects on anomaly detection, respectively. The input and optimization objective of the reconstruction task are as described in Section III-B. In the denoising task, we redesign the input part of the normal feature extractor. Its input is the forged anomaly sample, and the objective is the original samples without the contrast set. The two tasks share the same backbone.

The experimental results are shown in Table V. The results show that the anomaly classification performance decreases by 2% introducing the denoising task compared to the reconstruction task, and the anomaly localization in AUROC_{loc} and AP_{loc} decreases by 3% and 7%, respectively. Therefore, the reconstruction task is more suitable for NIGSF. The reason is that NIGSF relies on the guidance features as the prior knowledge to guide the segmentation module to locate anomalous regions. Therefore, it is important to ensure that the guidance features are noise-free normal features. The input of the reconstruction network is normal samples, while the input of the denoising network is forged anomalous samples. The features extracted from the reconstruction network are noise-free and more suitable for the guidance features. Although the denoising network can denoise some anomalous regions by training, it is difficult for a simple denoising network to guarantee a good denoising effect for all anomalous regions.

Therefore, the guidance features extracted from the denoising network may include noisy anomalous features, which cannot be used as guidance. In conclusion, the choice of the reconstruction task has the potential to achieve excellent anomaly detection performance.

4) *Comparison of the Proposed Method With Advanced Method Draem*: To demonstrate the performance and parameter advantages of the proposed NIGSF, we conduct three different settings compared with DRAEM.

- DRAEM₁: SOTA DRAEM with the large UNet network structure, *i.e.*, more decoders and encoders.
- DRAEM₂: SOTA DRAEM with the standard UNet network. The number of parameters is 1/3 less than that of DRAEM₁.
- NIGSF: The proposed method with the standard UNet network.

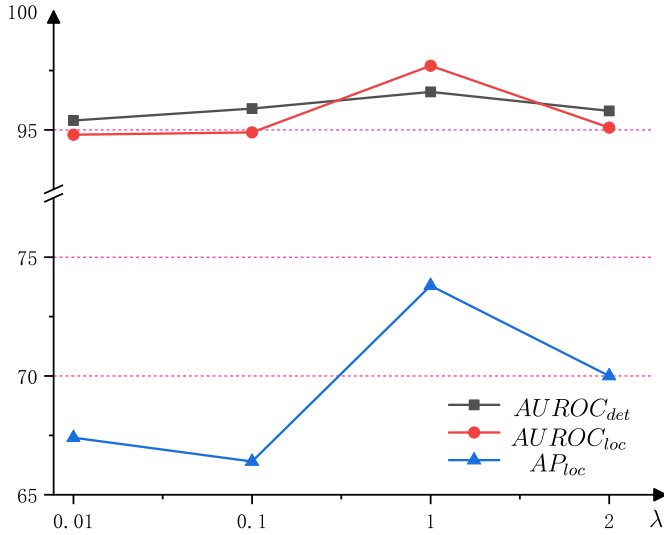
The experimental results are shown in Table VII. It shows that our method has better localization results than DRAEM₁, and the parameters are reduced by 1/3, which shows that NIGSF is more efficient and accurate. Second, after changing the network of DRAEM to the network structure of this paper, the anomaly classification and localization results are reduced by 3.7%, 2.3% and 12.9%, respectively. By comparing DRAEM₂ with the proposed method, it is shown that our method is more competitive. The smaller model brings more competitive localization and classification results.

5) *Contrast Set*: To validate the effect of the number of images n in the contrast set, ablation studies with different values of n are explored. The results are shown in Table VIII. “W/O.A.” indicates whether the contrast set contains the original image A. From Table VIII (2) and (3), it can be observed that the performance of anomaly detection containing A is lower than the performance without A. Including the original image in the contrast set makes the segmentation module in NIGSF overly focused on the guidance of low-level features, which produces a negative effect. It is impossible to find normal samples in the contrast set that completely overlap with the inference phase at low-level features, resulting in serious degradation of NIGSF performance. From the results of Table VIII (1), (2) and (4), the localization performance with $n = 4$ exceeds the remaining settings. Therefore, n in the contrast set is set to 4 without the original image. A large n leads to the more significant difference between the image in the contrast set and the input image, which makes the extracted guidance feature may be noisy. Noisy guidance features fail to provide an effective reference, which

TABLE VIII

THE IMPACT OF THE NUMBER OF SAMPLES n IN CONTRAST SET ON NIGSF. “W/O. A.” INDICATES WHETHER THE CONTRAST SET CONTAINS THE INPUT IMAGES. “- / - / -” REPRESENTS THE AVERAGE RESULT OF 5 TEXTURE CLASSES, THE AVERAGE RESULT OF 10 OBJECT CLASSES, AND THE OVERALL AVERAGE RESULT, RESPECTIVELY

	Setting		Metric		
	n	W/O. A.	$AUROC_{det}$	$AUROC_{loc}$	AP_{loc}
(1)	$n = 2$	✗	99.4 / 95.4 / 96.9	99.3 / 95.9 / 96.5	76.7 / 70.3 / 72.5
(2)	$n = 4$	✗	99.2 / 95.4 / 96.6	98.5 / 96.5 / 97.2	78.4 / 71.5 / 73.8
(3)	$n = 5$	✓	99.3 / 95.9 / 97.0	98.5 / 96.2 / 97.0	79.3 / 68.7 / 72.2
(4)	$n = 9$	✗	99.2 / 94.1 / 95.8	98.4 / 95.4 / 96.4	79.8 / 69.3 / 72.8

Fig. 8. Ablation study of weight parameter λ in loss functions.

results in the segmentation module’s localization effect being weakened. When n is too small, the segmentation module accepts less variation in the guidance features, which weakens the generalization ability of the segmentation module.

6) *Ablation Study of λ* : To validate the effect of hyperparameters λ in the loss function on the model, four sets of comparison experiments with hyperparameters λ were conducted. In the experiment, λ is set to 0.01, 0.1, 1, and 2, respectively. The experimental results are shown in Figure 8. When $\lambda = 0.01$, it indicates that the loss of the normal feature extractor is smaller than the overall loss. From the results in Figure 8, it is found that the performance of both localization and classification decreases, which may be due to the loss function causing the normal feature extractor not fully to converge. Normal feature extractor plays a critical role in this paper. If the reconstruction effect of the normal feature extractor is not significant, then there may be noise in the extracted guidance features, which affects the learning effect of the segmentation module. Therefore, the value of λ cannot be too small. When λ is large, the performance of the model starts to degrade. Therefore, the optimal hyperparameter $\lambda = 1$ is chosen in this paper.

V. CONCLUSION

In this paper, a novel unsupervised anomaly detection paradigm is proposed to introduce normal image as reference.

Specifically, based on the novel paradigm, a Normal Image Guided Segmentation Framework (NIGSF) is proposed, which consists of a Normal Guided Network (NGN) and a Saliency Augmentation Module (SAM). In NGN, we introduce the contrast set, which is a normal guidance image candidate set for input samples. From the guidance images, we extract normal features with unambiguous semantics and free from noise. Meanwhile, a straightforward and efficient feature fusion strategy is developed to fuse guidance features and features to be detected, which allows the segmentation module to efficiently detect “abnormal features” different from the guidance features. To facilitate the segmentation of “anomalies” by NIGSF, we introduce the SAM that generates forged anomaly samples utilizing saliency maps and random Perlin noise. Multiple experiments confirmed the efficacy of NIGSF, specifically its anomaly segmentation capability. In the future, we will gradually address the challenge of low accuracy of existing models in “transistor” category localization and propose more reasonable alignment methods to make the guidance features more useful.

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